# EXPERIMENT #72

Timestamp: 20260331_095049 | Environment: Local

## 1. Experiment Parameters & Network Statistics

| Parameter | Value |
|---|---|
| Number of sticks (N) | 30 |
| Stick length (a) | 0.6 |
| Domain size | 1.0 x 1.0 |
| Source region | x < 0.1 |
| Drain region | x > 0.9 |
| Total intersections | 118 |
| Has percolation | [OK] Yes |
| # Percolating paths | 1 |
| Nodes before pruning | 118 |
| Nodes after pruning | 118 |
| Edges before pruning | 206 |
| Edges after pruning | 206 |

## 2. Network Visualizations

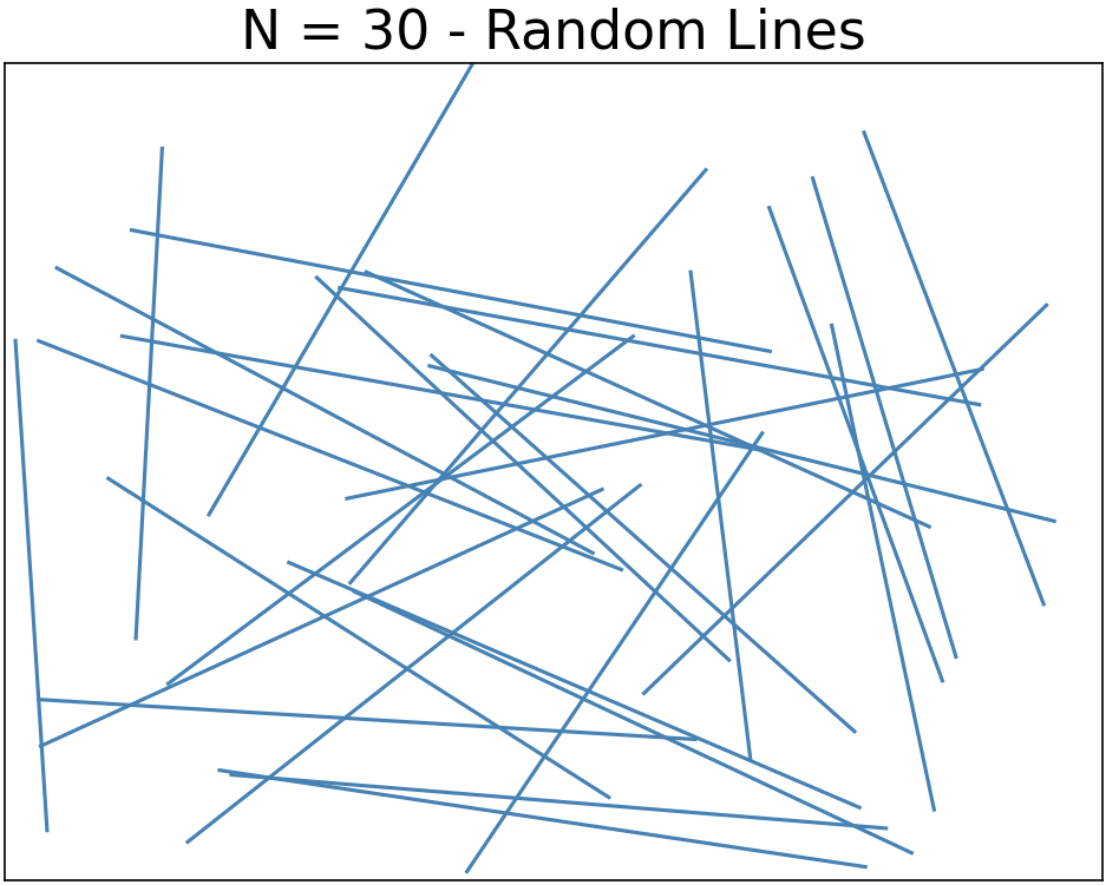



Fig. -- 01 random lines N30

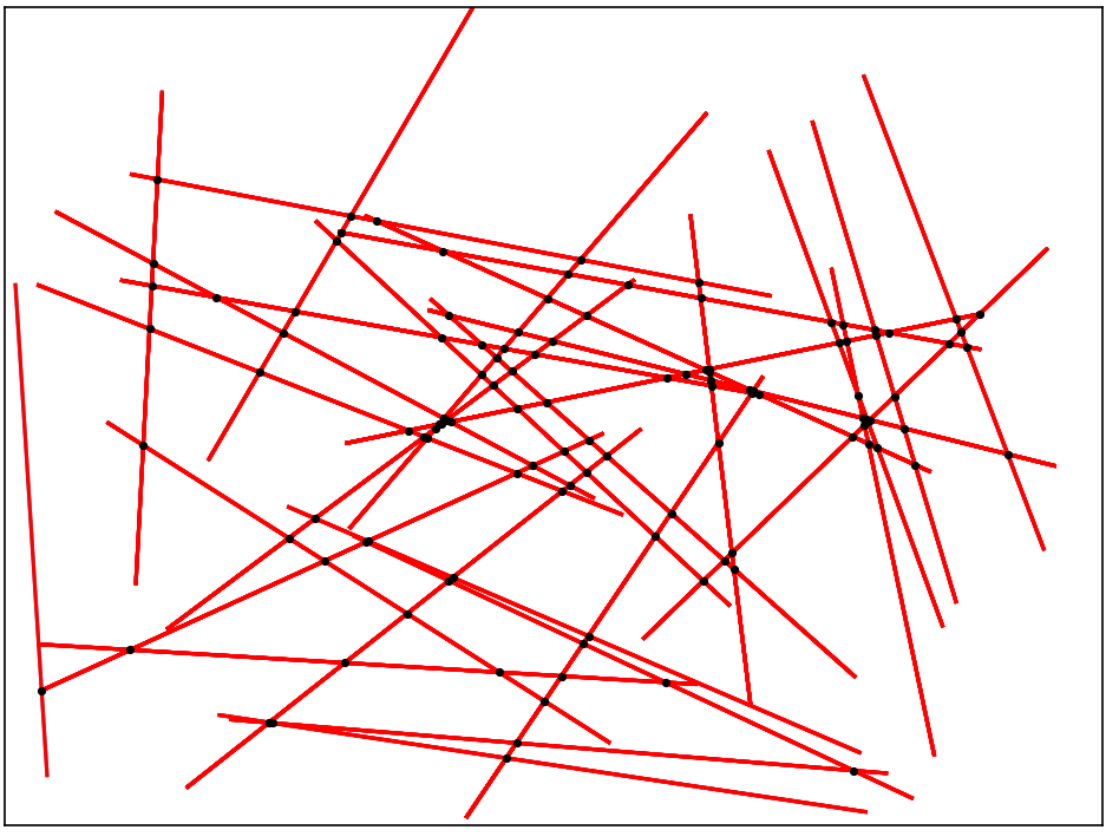



Fig. -- 02 intersections N30

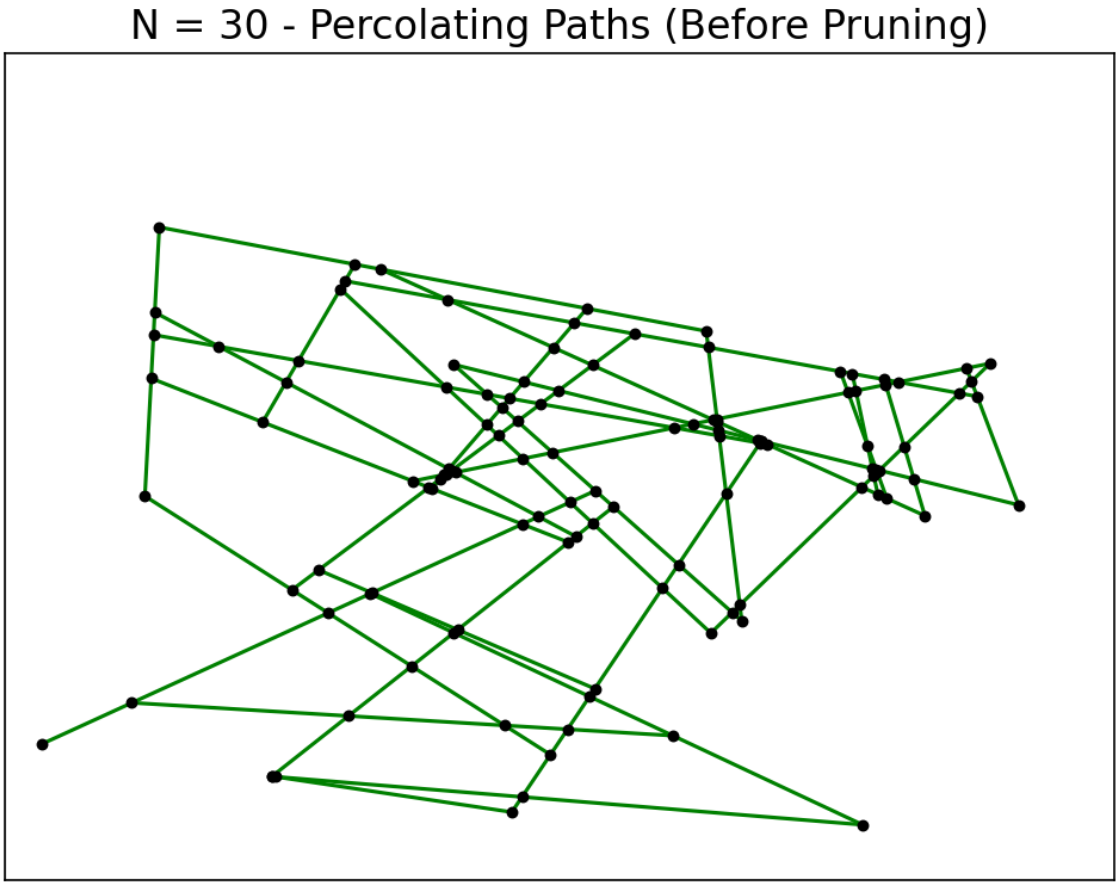



Fig. -- 03 percolating paths before pruning N30

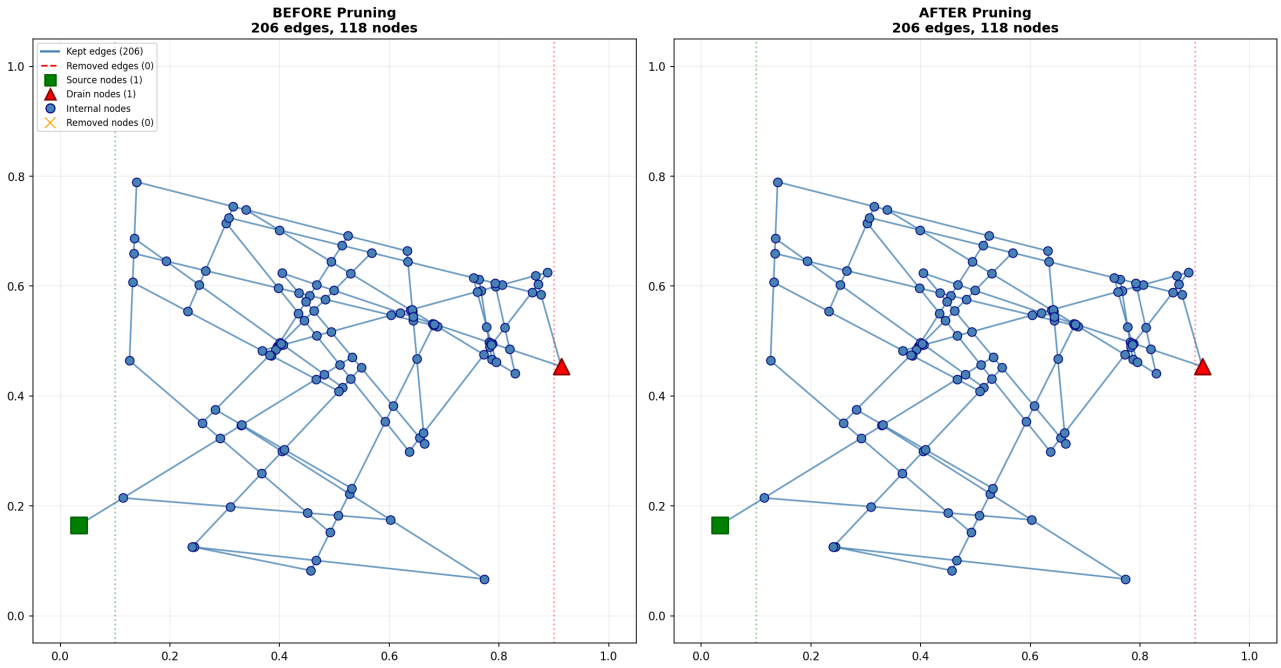



Fig. -- 04 before after pruning N30

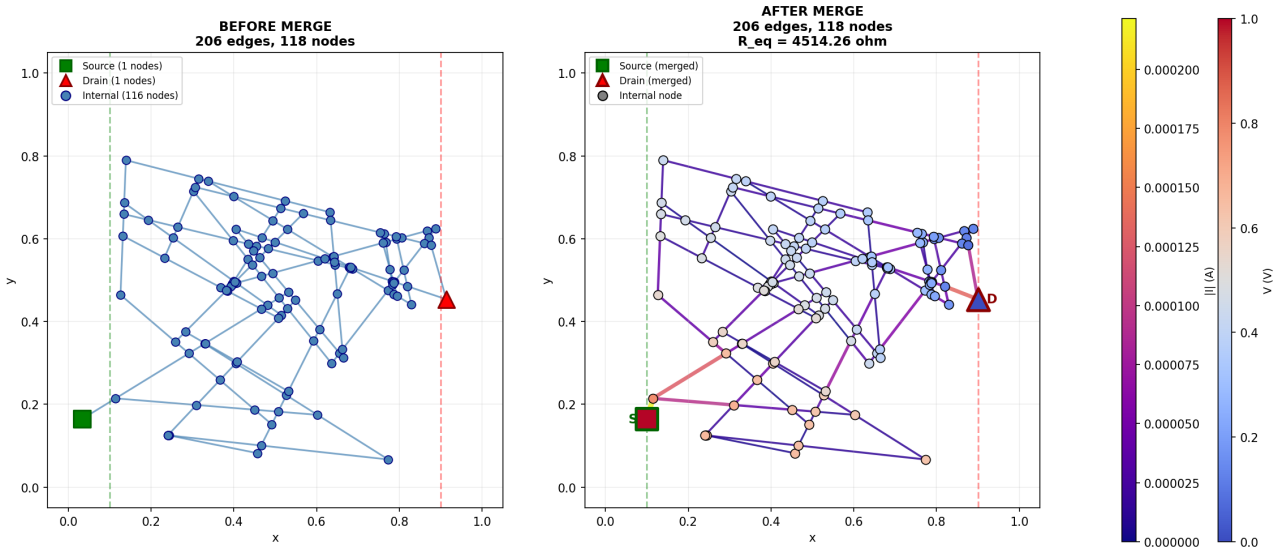



Fig. -- network merge explained

## 3. Edge Resistances (Pruned Network)

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 0 | 5 | 12 | 1.0000 | 1.0000 | 0.001133 |
| 1 | 12 | 2 | 1.0000 | 1.0000 | 0.007115 |
| 2 | 2 | 7 | 1.0000 | 1.0000 | 0.013635 |
| 3 | 7 | 1 | 1.0000 | 1.0000 | 0.017918 |
| 4 | 1 | 11 | 1.0000 | 1.0000 | 0.010158 |
| 5 | 11 | 3 | 1.0000 | 1.0000 | 0.006083 |
| 6 | 3 | 10 | 1.0000 | 1.0000 | 0.002067 |
| 7 | 10 | 0 | 1.0000 | 1.0000 | 0.001187 |
| 8 | 0 | 4 | 1.0000 | 1.0000 | 0.022502 |
| 9 | 4 | 6 | 1.0000 | 1.0000 | 0.041371 |
| 10 | 6 | 8 | 1.0000 | 1.0000 | 0.042250 |
| 11 | 8 | 9 | 1.0000 | 1.0000 | 0.046672 |
| 12 | 0 | 23 | 1.0000 | 1.0000 | 0.021315 |
| 13 | 23 | 25 | 1.0000 | 1.0000 | 0.016404 |
| 14 | 25 | 16 | 1.0000 | 1.0000 | 0.023174 |
| 15 | 16 | 21 | 1.0000 | 1.0000 | 0.010434 |
| 16 | 21 | 17 | 1.0000 | 1.0000 | 0.007329 |
| 17 | 17 | 22 | 1.0000 | 1.0000 | 0.008293 |
| 18 | 22 | 18 | 1.0000 | 1.0000 | 0.014693 |
| 19 | 18 | 24 | 1.0000 | 1.0000 | 0.011686 |
| 20 | 24 | 19 | 1.0000 | 1.0000 | 0.011368 |
| 21 | 19 | 20 | 1.0000 | 1.0000 | 0.011415 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 22 | 39 | 31 | 1.0000 | 1.0000 | 0.010529 |
| 23 | 31 | 37 | 1.0000 | 1.0000 | 0.004349 |
| 24 | 37 | 33 | 1.0000 | 1.0000 | 0.007871 |
| 25 | 33 | 32 | 1.0000 | 1.0000 | 0.017467 |
| 26 | 32 | 42 | 1.0000 | 1.0000 | 0.021952 |
| 27 | 42 | 16 | 1.0000 | 1.0000 | 0.034686 |
| 28 | 16 | 34 | 1.0000 | 1.0000 | 0.021946 |
| 29 | 34 | 1 | 1.0000 | 1.0000 | 0.019451 |
| 30 | 1 | 43 | 1.0000 | 1.0000 | 0.027211 |
| 31 | 43 | 40 | 1.0000 | 1.0000 | 0.048672 |
| 32 | 40 | 36 | 1.0000 | 1.0000 | 0.034453 |
| 33 | 36 | 41 | 1.0000 | 1.0000 | 0.048423 |
| 34 | 41 | 44 | 1.0000 | 1.0000 | 0.025728 |
| 35 | 44 | 38 | 1.0000 | 1.0000 | 0.026406 |
| 36 | 38 | 35 | 1.0000 | 1.0000 | 0.008802 |
| 37 | 48 | 52 | 1.0000 | 1.0000 | 0.003325 |
| 38 | 52 | 2 | 1.0000 | 1.0000 | 0.007077 |
| 39 | 2 | 47 | 1.0000 | 1.0000 | 0.013598 |
| 40 | 47 | 17 | 1.0000 | 1.0000 | 0.014009 |
| 41 | 17 | 51 | 1.0000 | 1.0000 | 0.013045 |
| 42 | 51 | 45 | 1.0000 | 1.0000 | 0.016821 |
| 43 | 45 | 46 | 1.0000 | 1.0000 | 0.020128 |
| 44 | 46 | 31 | 1.0000 | 1.0000 | 0.018568 |
| 45 | 31 | 49 | 1.0000 | 1.0000 | 0.012388 |
| 46 | 49 | 50 | 1.0000 | 1.0000 | 0.012860 |
| 47 | 74 | 47 | 1.0000 | 1.0000 | 0.084468 |
| 48 | 47 | 70 | 1.0000 | 1.0000 | 0.011563 |
| 49 | 70 | 34 | 1.0000 | 1.0000 | 0.014137 |
| 50 | 34 | 76 | 1.0000 | 1.0000 | 0.060665 |
| 51 | 76 | 3 | 1.0000 | 1.0000 | 0.096357 |
| 52 | 3 | 75 | 1.0000 | 1.0000 | 0.019060 |
| 53 | 75 | 69 | 1.0000 | 1.0000 | 0.086219 |
| 54 | 69 | 72 | 1.0000 | 1.0000 | 0.135563 |
| 55 | 72 | 68 | 1.0000 | 1.0000 | 0.057798 |
| 56 | 68 | 73 | 1.0000 | 1.0000 | 0.217468 |
| 57 | 73 | 71 | 1.0000 | 1.0000 | 0.171814 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 58 | 53 | 84 | 1.0000 | 1.0000 | 0.063698 |
| 59 | 84 | 85 | 1.0000 | 1.0000 | 0.234440 |
| 60 | 85 | 4 | 1.0000 | 1.0000 | 0.175078 |
| 61 | 4 | 80 | 1.0000 | 1.0000 | 0.186500 |
| 62 | 80 | 82 | 1.0000 | 1.0000 | 0.039507 |
| 63 | 82 | 68 | 1.0000 | 1.0000 | 0.027825 |
| 64 | 68 | 83 | 1.0000 | 1.0000 | 0.211776 |
| 65 | 83 | 86 | 1.0000 | 1.0000 | 0.008977 |
| 66 | 86 | 81 | 1.0000 | 1.0000 | 0.180784 |
| 67 | 81 | 35 | 1.0000 | 1.0000 | 0.098499 |
| 68 | 88 | 48 | 1.0000 | 1.0000 | 0.141275 |
| 69 | 48 | 5 | 1.0000 | 1.0000 | 0.018651 |
| 70 | 5 | 87 | 1.0000 | 1.0000 | 0.040628 |
| 71 | 87 | 77 | 1.0000 | 1.0000 | 0.144827 |
| 72 | 95 | 36 | 1.0000 | 1.0000 | 0.183516 |
| 73 | 36 | 94 | 1.0000 | 1.0000 | 0.266713 |
| 74 | 94 | 69 | 1.0000 | 1.0000 | 0.239777 |
| 75 | 69 | 80 | 1.0000 | 1.0000 | 0.145097 |
| 76 | 80 | 6 | 1.0000 | 1.0000 | 0.145129 |
| 77 | 98 | 100 | 1.0000 | 1.0000 | 0.028554 |
| 78 | 100 | 96 | 1.0000 | 1.0000 | 0.059851 |
| 79 | 96 | 49 | 1.0000 | 1.0000 | 0.085358 |
| 80 | 49 | 37 | 1.0000 | 1.0000 | 0.016737 |
| 81 | 37 | 61 | 1.0000 | 1.0000 | 0.013216 |
| 82 | 61 | 55 | 1.0000 | 1.0000 | 0.017558 |
| 83 | 55 | 97 | 1.0000 | 1.0000 | 0.086069 |
| 84 | 97 | 21 | 1.0000 | 1.0000 | 0.045301 |
| 85 | 21 | 70 | 1.0000 | 1.0000 | 0.018244 |
| 86 | 70 | 7 | 1.0000 | 1.0000 | 0.015670 |
| 87 | 7 | 99 | 1.0000 | 1.0000 | 0.284244 |
| 88 | 8 | 82 | 1.0000 | 1.0000 | 0.063372 |
| 89 | 82 | 72 | 1.0000 | 1.0000 | 0.029972 |
| 90 | 72 | 94 | 1.0000 | 1.0000 | 0.375339 |
| 91 | 94 | 40 | 1.0000 | 1.0000 | 0.232259 |
| 92 | 40 | 108 | 1.0000 | 1.0000 | 0.364926 |
| 93 | 109 | 41 | 1.0000 | 1.0000 | 0.228786 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 94 | 41 | 83 | 1.0000 | 1.0000 | 0.209370 |
| 95 | 83 | 73 | 1.0000 | 1.0000 | 0.005692 |
| 96 | 73 | 9 | 1.0000 | 1.0000 | 0.228594 |
| 97 | 15 | 110 | 1.0000 | 1.0000 | 0.261068 |
| 98 | 110 | 114 | 1.0000 | 1.0000 | 0.024856 |
| 99 | 114 | 66 | 1.0000 | 1.0000 | 0.211388 |
| 100 | 66 | 30 | 1.0000 | 1.0000 | 0.030239 |
| 101 | 30 | 115 | 1.0000 | 1.0000 | 0.262794 |
| 102 | 115 | 57 | 1.0000 | 1.0000 | 0.134316 |
| 103 | 57 | 111 | 1.0000 | 1.0000 | 0.084656 |
| 104 | 111 | 113 | 1.0000 | 1.0000 | 0.156488 |
| 105 | 113 | 23 | 1.0000 | 1.0000 | 0.181527 |
| 106 | 23 | 10 | 1.0000 | 1.0000 | 0.020128 |
| 107 | 10 | 75 | 1.0000 | 1.0000 | 0.021127 |
| 108 | 112 | 85 | 1.0000 | 1.0000 | 0.501654 |
| 109 | 85 | 113 | 1.0000 | 1.0000 | 0.400422 |
| 110 | 113 | 25 | 1.0000 | 1.0000 | 0.165123 |
| 111 | 25 | 76 | 1.0000 | 1.0000 | 0.061892 |
| 112 | 76 | 11 | 1.0000 | 1.0000 | 0.090274 |
| 113 | 11 | 43 | 1.0000 | 1.0000 | 0.017052 |
| 114 | 43 | 117 | 1.0000 | 1.0000 | 0.326194 |
| 115 | 117 | 88 | 1.0000 | 1.0000 | 0.515831 |
| 116 | 101 | 86 | 1.0000 | 1.0000 | 0.247493 |
| 117 | 86 | 44 | 1.0000 | 1.0000 | 0.244075 |
| 118 | 44 | 109 | 1.0000 | 1.0000 | 0.254514 |
| 119 | 109 | 95 | 1.0000 | 1.0000 | 0.003153 |
| 120 | 95 | 108 | 1.0000 | 1.0000 | 0.215863 |
| 121 | 108 | 117 | 1.0000 | 1.0000 | 0.642448 |
| 122 | 117 | 99 | 1.0000 | 1.0000 | 0.087079 |
| 123 | 99 | 52 | 1.0000 | 1.0000 | 0.290802 |
| 124 | 52 | 12 | 1.0000 | 1.0000 | 0.014193 |
| 125 | 12 | 116 | 1.0000 | 1.0000 | 0.607926 |
| 126 | 13 | 14 | 1.0000 | 1.0000 | 0.012025 |
| 127 | 14 | 15 | 1.0000 | 1.0000 | 0.006012 |
| 128 | 104 | 56 | 1.0000 | 1.0000 | 0.086910 |
| 129 | 56 | 62 | 1.0000 | 1.0000 | 0.024273 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 130 | 62 | 102 | 1.0000 | 1.0000 | 0.044772 |
| 131 | 102 | 106 | 1.0000 | 1.0000 | 0.160825 |
| 132 | 106 | 28 | 1.0000 | 1.0000 | 0.314195 |
| 133 | 28 | 105 | 1.0000 | 1.0000 | 0.113839 |
| 134 | 105 | 65 | 1.0000 | 1.0000 | 0.200476 |
| 135 | 65 | 13 | 1.0000 | 1.0000 | 0.018037 |
| 136 | 13 | 103 | 1.0000 | 1.0000 | 0.021198 |
| 137 | 103 | 14 | 1.0000 | 1.0000 | 0.009173 |
| 138 | 14 | 110 | 1.0000 | 1.0000 | 0.267081 |
| 139 | 110 | 29 | 1.0000 | 1.0000 | 0.241484 |
| 140 | 58 | 18 | 1.0000 | 1.0000 | 0.003872 |
| 141 | 18 | 45 | 1.0000 | 1.0000 | 0.006880 |
| 142 | 45 | 55 | 1.0000 | 1.0000 | 0.003573 |
| 143 | 55 | 32 | 1.0000 | 1.0000 | 0.005436 |
| 144 | 32 | 54 | 1.0000 | 1.0000 | 0.009920 |
| 145 | 54 | 56 | 1.0000 | 1.0000 | 0.003666 |
| 146 | 56 | 57 | 1.0000 | 1.0000 | 0.000108 |
| 147 | 57 | 53 | 1.0000 | 1.0000 | 0.030452 |
| 148 | 62 | 60 | 1.0000 | 1.0000 | 0.005450 |
| 149 | 60 | 33 | 1.0000 | 1.0000 | 0.014942 |
| 150 | 33 | 61 | 1.0000 | 1.0000 | 0.005345 |
| 151 | 61 | 46 | 1.0000 | 1.0000 | 0.001003 |
| 152 | 46 | 63 | 1.0000 | 1.0000 | 0.000556 |
| 153 | 63 | 19 | 1.0000 | 1.0000 | 0.004510 |
| 154 | 19 | 59 | 1.0000 | 1.0000 | 0.004557 |
| 155 | 77 | 59 | 1.0000 | 1.0000 | 0.043358 |
| 156 | 59 | 20 | 1.0000 | 1.0000 | 0.015972 |
| 157 | 20 | 78 | 1.0000 | 1.0000 | 0.068119 |
| 158 | 78 | 79 | 1.0000 | 1.0000 | 0.267226 |
| 159 | 74 | 22 | 1.0000 | 1.0000 | 0.106771 |
| 160 | 22 | 51 | 1.0000 | 1.0000 | 0.004751 |
| 161 | 51 | 97 | 1.0000 | 1.0000 | 0.065674 |
| 162 | 97 | 42 | 1.0000 | 1.0000 | 0.069553 |
| 163 | 42 | 91 | 1.0000 | 1.0000 | 0.124588 |
| 164 | 91 | 104 | 1.0000 | 1.0000 | 0.002139 |
| 165 | 104 | 111 | 1.0000 | 1.0000 | 0.171674 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 166 | 111 | 84 | 1.0000 | 1.0000 | 0.009494 |
| 167 | 84 | 112 | 1.0000 | 1.0000 | 0.267214 |
| 168 | 87 | 116 | 1.0000 | 1.0000 | 0.647420 |
| 169 | 116 | 58 | 1.0000 | 1.0000 | 0.568320 |
| 170 | 58 | 24 | 1.0000 | 1.0000 | 0.015559 |
| 171 | 24 | 63 | 1.0000 | 1.0000 | 0.015877 |
| 172 | 63 | 107 | 1.0000 | 1.0000 | 0.260933 |
| 173 | 29 | 26 | 1.0000 | 1.0000 | 0.013572 |
| 174 | 26 | 30 | 1.0000 | 1.0000 | 0.021906 |
| 175 | 30 | 28 | 1.0000 | 1.0000 | 0.004501 |
| 176 | 28 | 27 | 1.0000 | 1.0000 | 0.003061 |
| 177 | 26 | 66 | 1.0000 | 1.0000 | 0.008334 |
| 178 | 66 | 67 | 1.0000 | 1.0000 | 0.024227 |
| 179 | 67 | 65 | 1.0000 | 1.0000 | 0.036671 |
| 180 | 65 | 64 | 1.0000 | 1.0000 | 0.093944 |
| 181 | 91 | 54 | 1.0000 | 1.0000 | 0.092715 |
| 182 | 54 | 60 | 1.0000 | 1.0000 | 0.022489 |
| 183 | 60 | 90 | 1.0000 | 1.0000 | 0.167053 |
| 184 | 90 | 93 | 1.0000 | 1.0000 | 0.407918 |
| 185 | 93 | 27 | 1.0000 | 1.0000 | 0.357974 |
| 186 | 27 | 92 | 1.0000 | 1.0000 | 0.343837 |
| 187 | 92 | 64 | 1.0000 | 1.0000 | 0.521357 |
| 188 | 64 | 89 | 1.0000 | 1.0000 | 0.333354 |
| 189 | 71 | 101 | 1.0000 | 1.0000 | 0.422592 |
| 190 | 101 | 81 | 1.0000 | 1.0000 | 0.428277 |
| 191 | 81 | 38 | 1.0000 | 1.0000 | 0.089697 |
| 192 | 102 | 90 | 1.0000 | 1.0000 | 0.206375 |
| 193 | 90 | 50 | 1.0000 | 1.0000 | 0.144527 |
| 194 | 50 | 96 | 1.0000 | 1.0000 | 0.072497 |
| 195 | 96 | 39 | 1.0000 | 1.0000 | 0.087217 |
| 196 | 39 | 107 | 1.0000 | 1.0000 | 0.289474 |
| 197 | 107 | 78 | 1.0000 | 1.0000 | 0.335958 |
| 198 | 114 | 67 | 1.0000 | 1.0000 | 0.235615 |
| 199 | 67 | 105 | 1.0000 | 1.0000 | 0.163804 |
| 200 | 105 | 92 | 1.0000 | 1.0000 | 0.226937 |
| 201 | 92 | 98 | 1.0000 | 1.0000 | 0.290455 |

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ohm) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 202 | 98 | 79 | 1.0000 | 1.0000 | 0.276663 |
| 203 | 100 | 93 | 1.0000 | 1.0000 | 0.276038 |
| 204 | 93 | 106 | 1.0000 | 1.0000 | 0.040719 |
| 205 | 106 | 115 | 1.0000 | 1.0000 | 0.046900 |

## 4. Impedance Calculation

| Quantity | Value |
|---|---|
| Applied voltage (V) | 1.00 V |
| Resistance per segment k | 1 Ohm |
| Calculated impedance R_eq | 4514.255418 Ohm |
| Total current I | -- |

## Impedance for different k (resistance per segment) values

| k (Ohm/segment) | R_equivalent (Ohm) |
|---|---|
| 1 | 4.514255 |
| 10 | 45.142554 |
| 100 | 451.425542 |
| 1000 | 4514.255418 |
| 10000 | 45142.554177 |